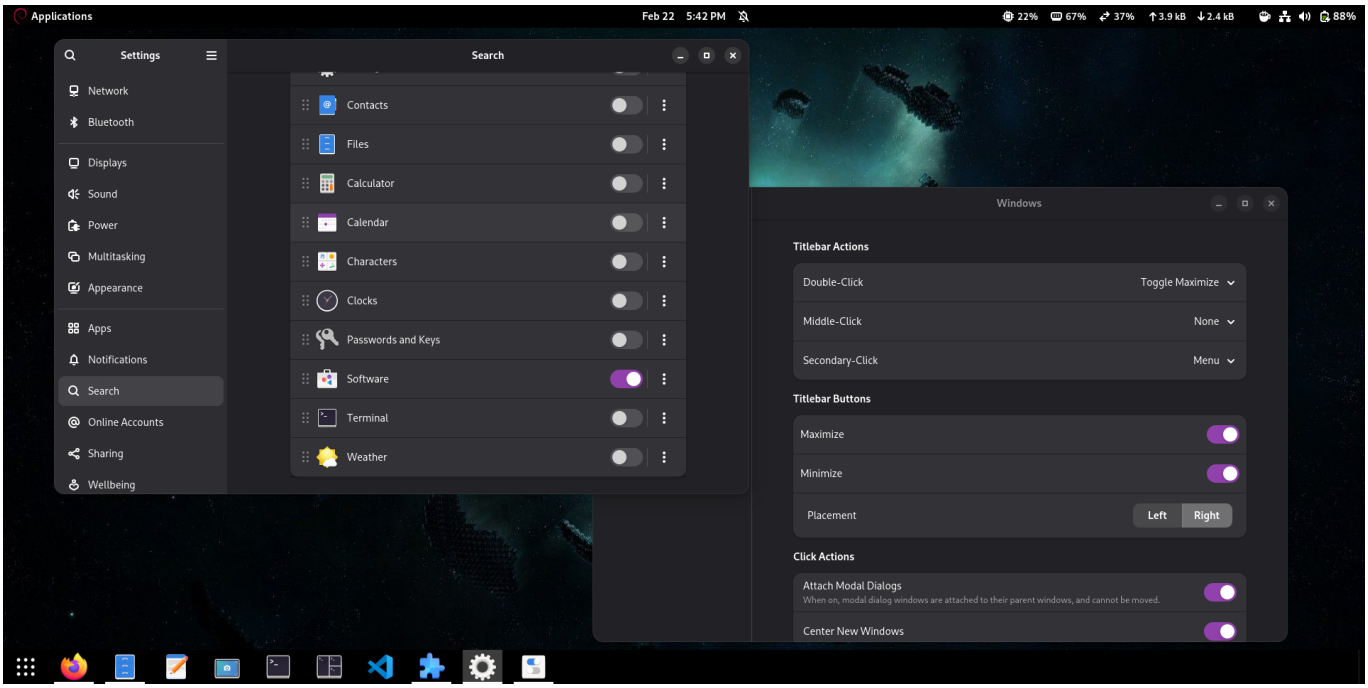
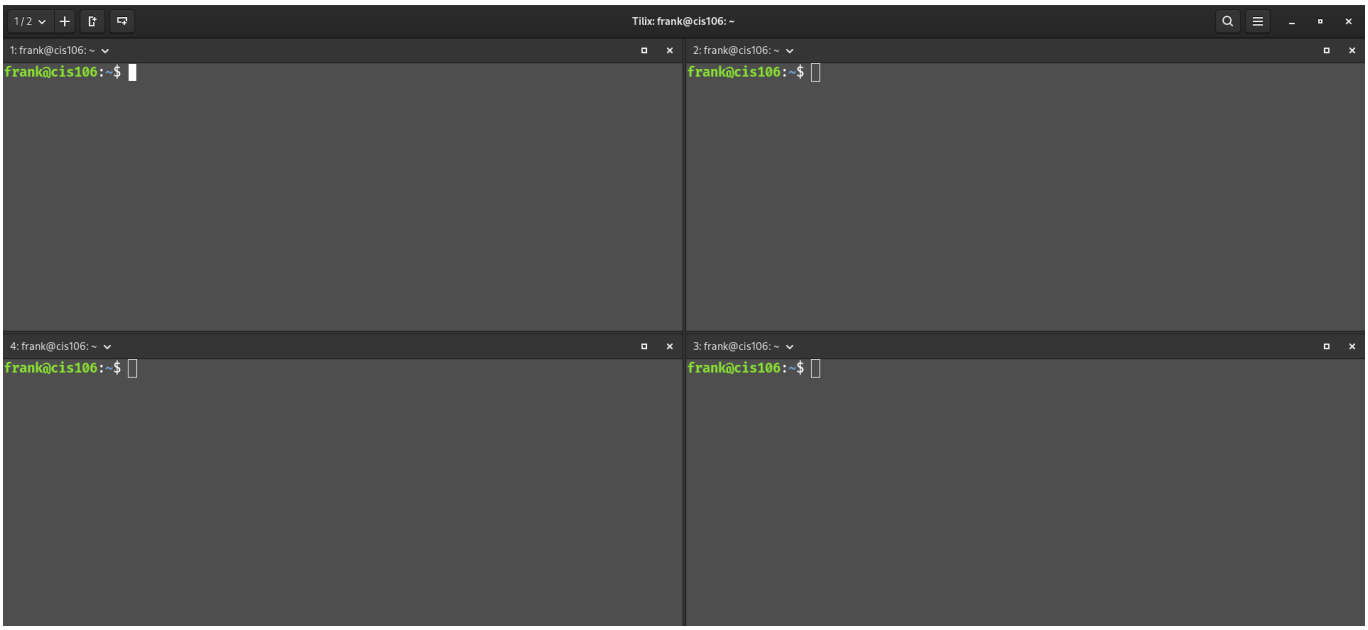
Lab 3 Submission

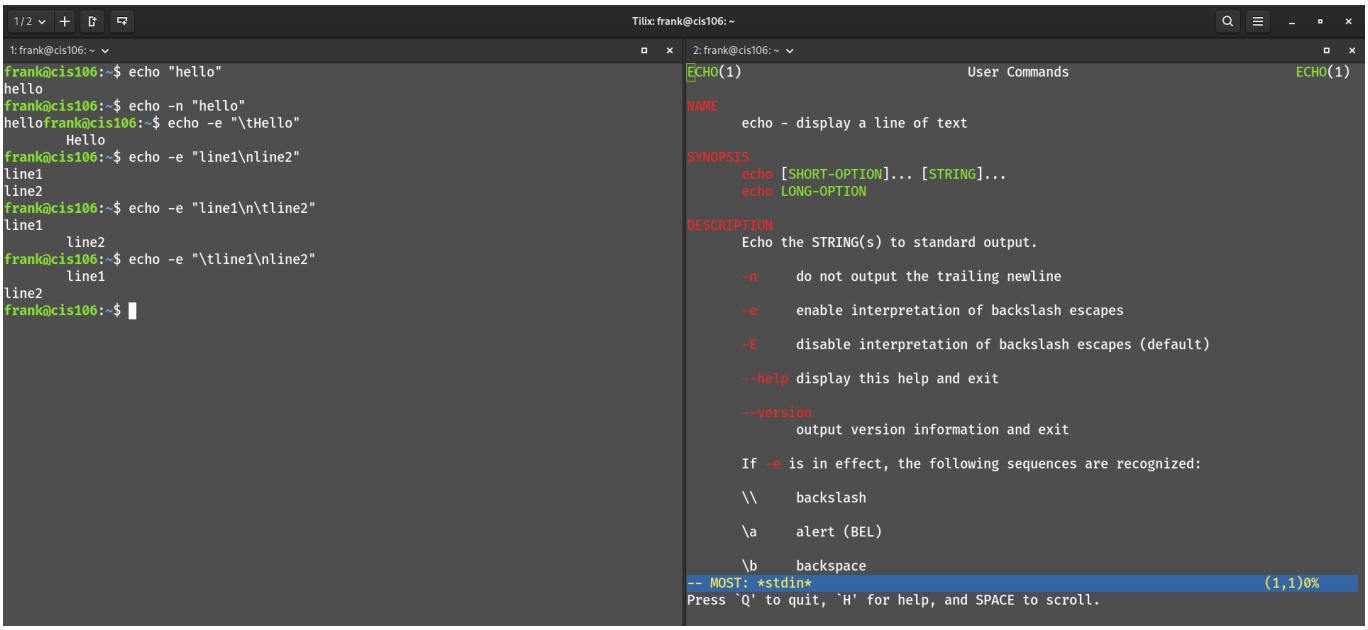
Question 1



Question 2



Question 3



Challenge Question

```
frank@cis106:~$ free -h -L
SwapUse         32Mi CachUse        1.3Gi  MemUse        1.3Gi MemFree        1.5Gi
frank@cis106:~$
```

```
       --peta Display the amount of memory in petabytes. Implies --si.

       -h, --human
              Show  all  output  fields  automatically scaled to shortest three digit
              unit and display the units of print out.  Following units are used.

                B = bytes
                Ki = kibibyte
                Mi = mebibyte
                Gi = gibibyte
                Ti = tebibyte
                Pi = pebibyte

              If unit is missing, and you have exbibyte of RAM or swap, the number is
              in tebibytes and columns might not be aligned with header.

       -w, --wide
              Switch to the wide mode. The wide mode produces lines  longer  than  80
              characters. In this mode buffers and cache are reported in two separate
              columns.

       -c, --count count
              Display the result count times.  Requires the -s option.

       -l, --lohi
              Show detailed low and high memory statistics.

       -L, --line
              Show  output  on  a  single line, often used with the -s option to show
              memory statistics repeatedly.

       -s, --seconds delay
              Continuously display the result delay  seconds apart.  You may actually
              specify any floating point number for delay using either  .  or  ,  for
              decimal  point.  usleep(3)  is  used  for microsecond resolution delay
-- MOST: *stdin*                                                          (65,1)49%
Press `Q' to quit, `H' for help, and SPACE to scroll.
```

```
frank@cis106:~$ uname -s -r -v -o
Linux 6.12.73+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.73-1 (2026-02-17) GNU/Linu
x
frank@cis106:~$
```

```
DESCRIPTION
       Print certain system information.  With no OPTION, same as -s.

       -a, --all
              print all information, in the following order, except omit -p and -i if
              unknown:

       -s, --kernel-name
              print the kernel name

       -n, --nodename
              print the network node hostname

       -r, --kernel-release
              print the kernel release

       -v, --kernel-version
              print the kernel version

       -m, --machine
              print the machine hardware name

       -p, --processor
              print the processor type (non-portable)

       -i, --hardware-platform
              print the hardware platform (non-portable)

       -o, --operating-system
              print the operating system

       --help display this help and exit

       --version
              output version information and exit

-- MOST: *stdin*                                                          (9,1)9%
Press `Q' to quit, `H' for help, and SPACE to scroll.
```

```
frank@cis106:~$ date --rfc-3339 ns
2026-02-22 18:18:37.944329139-05:00
frank@cis106:~$
```

```
       -f, --file=DATEFILE
              like --date; once for each line of DATEFILE

       -I[FMT], --iso-8601[=FMT]
              output date/time in ISO 8601 format.  FMT='date' for date only (the de-
              fault), 'hours', 'minutes', 'seconds', or 'ns' for date and time to the
              indicated precision.  Example: 2006-08-14T02:34:56-06:00

       --resolution
              output the available resolution of timestamps Example: 0.000000001

       -R, --rfc-email
              output date and time in RFC 5322 format.  Example:  Mon,  14  Aug  2006
              02:34:56 -0600

       --rfc-3339=FMT
              output  date/time  in  RFC 3339 format.  FMT='date', 'seconds', or 'ns'
              for date and time to  the  indicated  precision.   Example:  2006-08-14
              02:34:56-06:00

       -r, --reference=FILE
              display the last modification time of FILE

       -s, --set=STRING
              set time described by STRING

       -u, --utc, --universal
              print or set Coordinated Universal Time (UTC)

       --help display this help and exit

       --version
              output version information and exit

       All  options  that  specify the date to display are mutually exclusive.  I.e.:
-- MOST: *stdin*                                                          (21,1)11%
Press `Q' to quit, `H' for help, and SPACE to scroll.
```